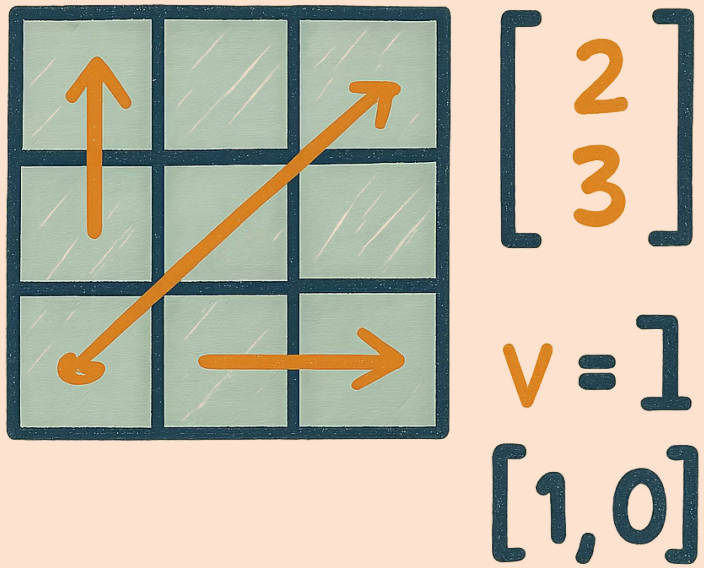
Linear Algebra — II

Foundations for B.Tech. Engineering Mathematics

B.TECH. MATHEMATICS

SEMESTER II



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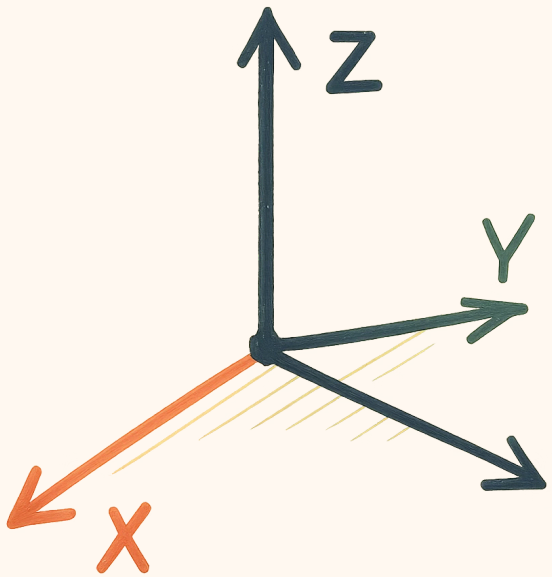
CHAPTER 1

Recap: Vector Spaces & Subspaces



Vector Space

A set **V** closed under vector addition and scalar multiplication, satisfying 8 axioms (associativity, commutativity, identity, inverses, distributivity).

Subspace Criteria

A subset $\mathbf{W} \subseteq \mathbf{V}$ is a subspace if:

- Zero vector $\mathbf{0} \in \mathbf{W}$
- Closed under addition: $\mathbf{u}, \mathbf{v} \in \mathbf{W} \Rightarrow \mathbf{u}+\mathbf{v} \in \mathbf{W}$
- Closed under scalar multiplication: $\alpha\mathbf{u} \in \mathbf{W}$

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CHAPTER 2

Linear Transformations

A map **T: V → W** is linear if it preserves vector addition and scalar multiplication.

Definition

T(u + v) = T(u) + T(v) and **T(αu) = αT(u)** for all u, v ∈ V and scalar α.

Key Properties

Maps zero to zero. Preserves linear combinations. Kernel and Range are subspaces.

Examples

Rotation, reflection, projection in $\mathbb{R}^n$; differentiation operator on polynomial spaces.

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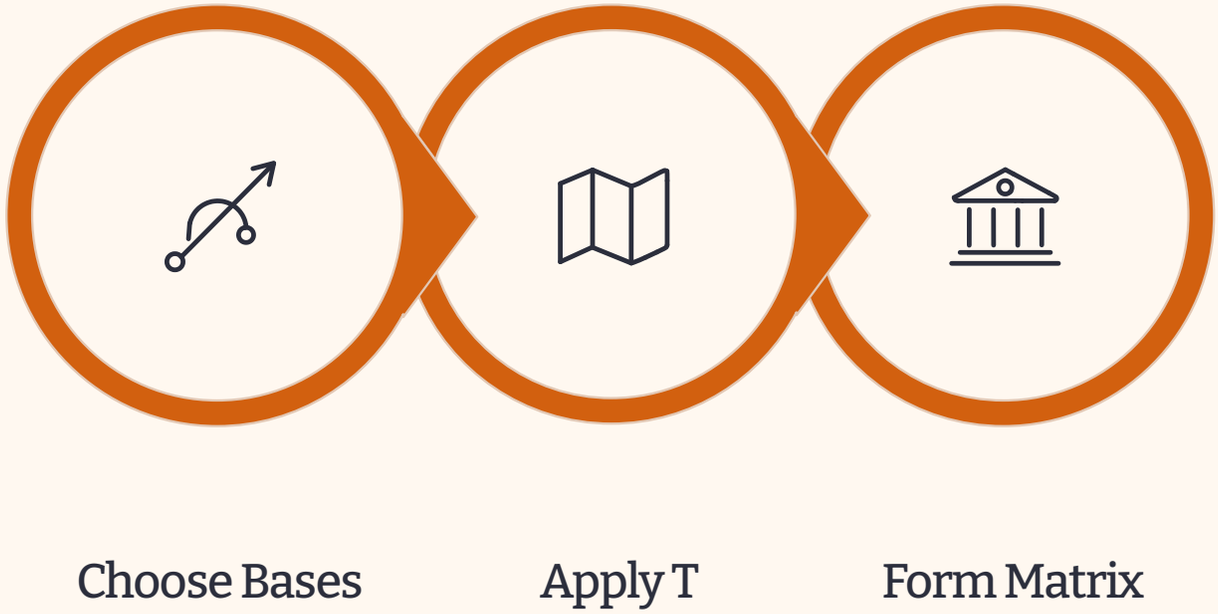
CHAPTER 3

Matrix Representation of Linear Transformations



Every linear transformation **T:** $\mathbb{R}^n \to \mathbb{R}^m$ corresponds uniquely to an **m × n** matrix **A** such that **T(x) = Ax**. The j-th column of A is $\mathbf{T(e_j)}$ expressed in the target basis.

Key Insight: Composition of transformations corresponds to matrix multiplication — $\mathbf{T_2}$ $\mathbf{T_1}$ maps to $\mathbf{A_2A_1}$.

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CHAPTER 4

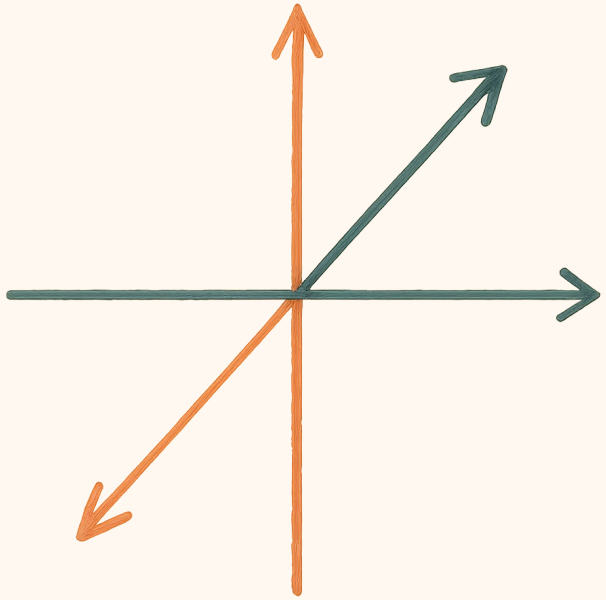
Eigenvalues & Eigenvectors

The Core Equation

For a square matrix **A**, a non-zero vector **v** is an eigenvector if:

$$Av = \lambda v$$

where **λ** is the eigenvalue. The vector's direction is *unchanged* by the transformation — only scaled.



Finding Eigenvalues

Solve the **characteristic equation**:

$$\det(A - \lambda I) = 0$$

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CHAPTER 5

Diagonalization of Matrices

An **n × n** matrix A is **diagonalizable** if $\mathbf{A = PDP^{-1}}$, where D is diagonal (eigenvalues) and P contains eigenvectors.

Condition

A has **n linearly independent** eigenvectors.

Why It Matters

$\mathbf{A^k = PD^kP^{-1}}$ — computing powers becomes trivial.

Engineering Use

Solving systems of differential equations, stability analysis, vibration modes.

⚠ **Note:** Not all matrices are diagonalizable. Symmetric matrices always are.

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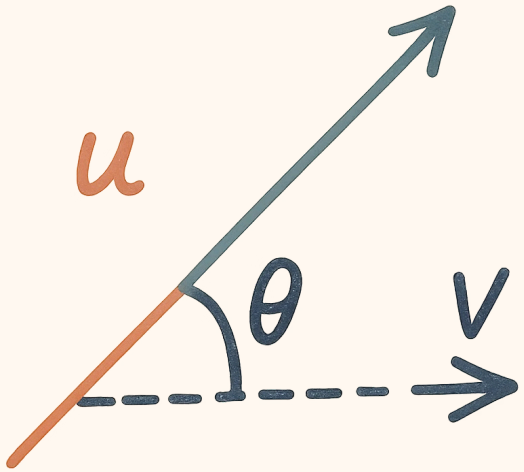
CHAPTER 6

Inner Product Spaces



Beyond the Dot Product

An **inner product** $\langle u, v \rangle$ generalizes the dot product to abstract vector spaces, satisfying:

- **Linearity:** $\langle \alpha u + v, w \rangle = \alpha \langle u, w \rangle + \langle v, w \rangle$
- **Symmetry:** $\langle u, v \rangle = \langle v, u \rangle$
- **Positive-definiteness:** $\langle v, v \rangle \geq 0$, equals 0 iff $v = 0$

Induces a **norm**: $\| v \| = \sqrt{\langle v, v \rangle}$ and a notion of **angle** between vectors.

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CHAPTER 7

Orthogonality & Gram-Schmidt Process

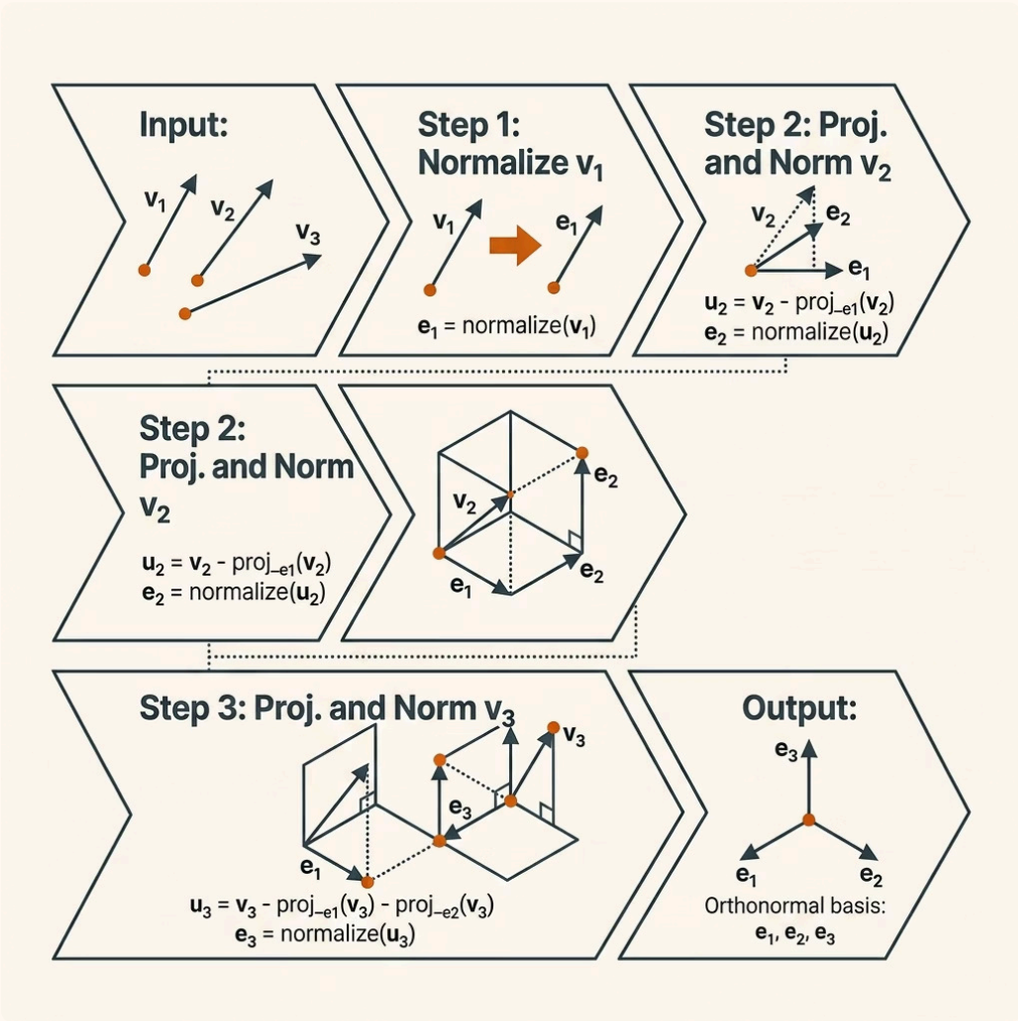
Orthogonality

Vectors **u, v** are orthogonal if **u, v = 0**. An **orthonormal set** satisfies $e_i, e_j = \delta_{ij}$.

Gram-Schmidt Algorithm

Converts any basis $\{v_1, \ldots, v_n\}$ into an orthonormal basis $\{e_1, \ldots, e_n\}$:

$$u_k = v_k - \sum_{j=1}^{k-1} \frac{\langle v_k, e_j \rangle}{\langle e_j, e_j \rangle} e_j, \qquad e_k = \frac{u_k}{\|u_k\|}$$



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CHAPTER 8 — APPLICATIONS

Linear Algebra in Engineering

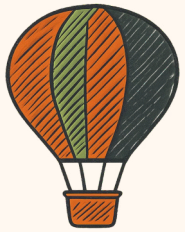
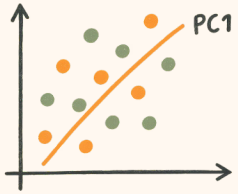


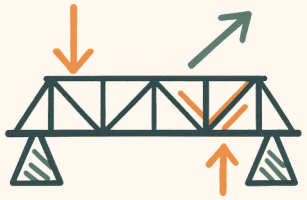
Image Processing

Images are matrices. Compression (JPEG), filtering, and transformations rely on SVD and eigendecomposition.



Data Science & ML

PCA uses eigenvectors of the covariance matrix to reduce dimensionality while preserving variance.



Structural Engineering

System of linear equations models forces in trusses. Eigenvalues reveal natural frequencies and resonance modes.

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Q&A & Key Takeaways

01

Vector Spaces

Foundation of all linear structures — understand subspaces and basis.

02

Transformations & Matrices

Every linear map has a matrix; composition = multiplication.

03

Eigenvalues & Diagonalization

Simplify powers, solve ODEs, analyze stability.

04

Inner Products & Orthogonality

Gram-Schmidt builds orthonormal bases — critical for numerical stability.

Next Steps: Practice eigenvalue problems, explore SVD, and apply concepts to your domain projects.

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